

Lorenzo Liuzzo

PHYSICIST · SOFTWARE ENGINEER · AI SYSTEMS

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Physicist and software engineer specialized in AI. Physics taught me to derive the model before writing the code; lately most of my time goes into learning how much of that writing can be delegated to a model, and where a person still has to decide.

Experience

MyThingsLab

 MyThingsLab 

PERSONAL PROJECT

July 2026 - Present

- **A harness, not a single tool.** Started in July 2026 to see how far AI-assisted development can go, and where it should stop: **more than 50 small Python tools** in the first month, each one independent, all sharing one SDK — **my-things-core**  — and its five contracts (ledger, policy, engine, GitHub, isolation), so nothing gets reimplemented twice.
- **Control, not autonomy.** Headless workers pick up an issue, do the deterministic pre-work with no model call, ask the LLM once for the step that needs judgement, and close it as a pull request — never a merge, so the last word stays with a person.
- **No model inside the core.** The SDK is dependency-free and shells out to **gh** and **git**, and the LLM sits behind a single **Engine** protocol whose deterministic default lets the whole fleet be exercised and tested without a token spent.

Money Balling AI

 MoneyBallingAI 

FOUNDER AND TECHNICAL LEAD

September 2025 - Present

- A team of four building one model that predicts the **exact box score** of an NBA game — every player and team line — with calibrated uncertainty. I am responsible for the architecture, from raw play-by-play to a calibrated prediction:
- **Data** — play-by-play parsed into periods, lineup stints and possessions, aggregated into cumulative stat windows and modelled as heterogeneous stint graphs, with stable global identities for players and teams across seasons.
- **Encoder** — a heterogeneous GNN over those graphs with per-entity cross-game and hierarchical season/game/stint memory, pretrained on a box-score delta task for stable player and team strength embeddings.
- **Simulator** — a vectorized Monte Carlo possession engine whose policies are behavior-cloned on real possessions; **calibration** scores per-stat distributions against the actual box score with MAE, CRPS, PIT and coverage, test-first and under CI.

Private Tutoring in Mathematics and Physics

Milan, Italy

IN-HOME TUTORING

2020 - Present

Continuous support for 20+ students: exam preparation, remedial tutoring and periodic reports to families.

Barista

Milan, Italy

DON SALVATORE - PIZZAIUOLO E OSTE

September - October 2024

Waiter

Milan, Italy

BAR TERRAZZA CLÉR

May - July 2022

Scout Leader

Milan, Italy

REPARTO BREITHORN - GRUPPO MI8 - AGESCI

2021 - 2022

Educational activities for youth aged 12-16: camps, outings, coordination with educators and families.

Education

University of Milan, Milan-Bicocca and Pavia

Milan/Pavia, Italy

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE FOR SCIENCE AND TECHNOLOGY

September 2025 - Present

University of Milan

Milan, Italy

BACHELOR'S DEGREE IN PHYSICS, GRADE: 100/110

September 2021 - July 2025

Thesis: *ray tracing and Monte Carlo methods for uncertainty quantification in radio-telescope pointing for CMB observations.*

I.I.S. Severi-Correnti

Milan, Italy

SCIENTIFIC DIPLOMA, GRADE: 100/100

September 2016 - June 2021

EF International Language School

Brighton, England

FULL-IMMERSION INTENSIVE ENGLISH LANGUAGE COURSE

June - September 2019

Schools & Workshops

School on Quantum Simulation

DEPARTMENT OF PHYSICS, UNIVERSITY OF MILAN

Milan, Italy

September 2026

Qiskit Fall Fest

IBM QUANTUM · DEPARTMENT OF PHYSICS, UNIVERSITY OF MILAN

Milan, Italy

October - November 2025

Projects

Gauge-Equivariant Graph Neural Networks

 [lorenzoliuzzo/EGNN](#)

DESIGNER/DEVELOPER

July - August 2026

Lattice Standard Model field theory simulated with gauge-equivariant GNNs in **PyTorch Geometric**: $SU(3) \times SU(2) \times U(1)$ gauge groups, Wilson-Dirac fermions and Higgs spontaneous symmetry breaking.

MSc Coursework Projects

DESIGNER/DEVELOPER

May - July 2026

- **Supervised Learning on Food Images** — a **PyTorch CNN** under 10M parameters classifying the 251 classes of the FoodX-251 dataset.
- **Transverse Field Ising Model** — quantum phase transitions explored on **PennyLane**.
- **Unsupervised Learning on Country Data** — clustering and dimensionality reduction on socio-economic country-level data.

Telescope Pointing Simulation

 [lorenzoliuzzo/thesis-L30](#)

DESIGNER/DEVELOPER

June - July 2025

Julia model of a telescope's reflective system, propagating mirror defects to the pointing direction via *ray tracing* and *Monte Carlo* simulation.

Systems & Numerical Programming

DESIGNER/DEVELOPER

2023 - 2024

- **Ray Tracing Engine** — pair project at UNIMI: a physical model turned into structured, tested and documented numerical code, while rapidly learning a new language (**Nim**).
- **Dimensional Analysis Library** — “CTDA: Compile Time Dimensional Analysis”, C++ numeric types carrying units of measurement in the type system.

How I Build Software

- **AI across the whole workflow, not only the coding step.** Composable tools cover research, design, implementation, testing and documentation, so the model is used wherever it earns its place rather than at a single step.
- **Deterministic first, models only where judgement is needed.** The LLM sits behind a single **Engine** seam, so a tool's entire skeleton runs and is tested at zero token cost.
- **Test-driven, and always through a pull request.** Tests are written with the code and often before it; **pytest**, coverage and **ruff** run on every push, and work lands by review rather than by pushing to a main branch.
- **Every side effect passes a gate.** A policy engine rules each action allow/ask/deny, tools open pull requests rather than merging, and every **ask** reaches me over Telegram before anything happens.
- **Understand the problem before writing the code.** A habit kept from physics: work out the model first, decide what the code actually has to compute, then write it — typed, documented, and with no abstraction built for a need that does not exist yet.

Technical Skills

Languages

Italian, English C1

Programming Languages

Python, C++, Julia, C, Nim

AI Engineering

Claude, Claude Code, Agentic workflows, Prompt & context engineering

ML & Data Science

PyTorch, PyTorch Geometric, Scikit-learn, Pandas, NumPy, SciPy, Jupyter

Databases & Quantum

Neo4j, Cypher, PennyLane, Qiskit

Practices

Test-driven development, Pull request workflow, CI/CD, Static typing

Development Tools

Git, GitHub, GitHub Actions, pytest, ruff

Typesetting

LaTeX, Typst, Markdown, Jekyll

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